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Studierichting: Informatica
Oefeningenreeks 5F nr. 1.1

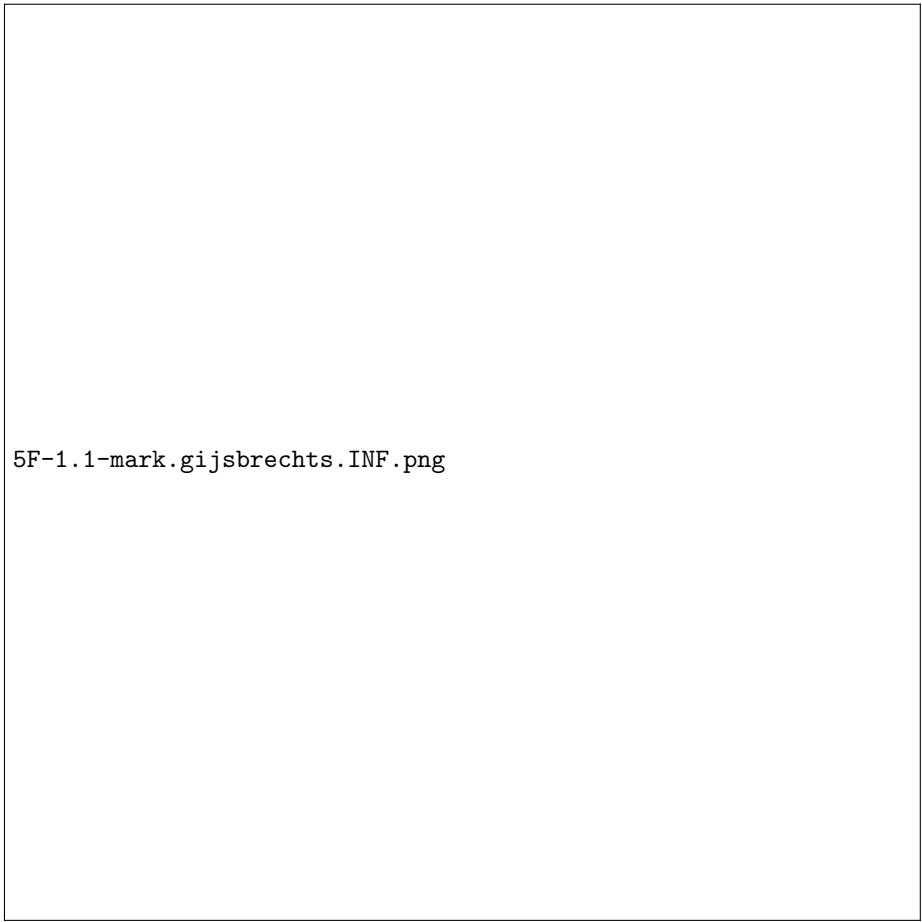
Bepaal van de volgende functie gegeven in poolcoördinaten de oppervlakte begrensd door de grafiek.

$$r = -\cos \theta$$

r is positief op interval $\left[\frac{\pi}{2}, \frac{3\pi}{2}\right]$

$$\begin{aligned} S &= \frac{1}{2} \int_{\frac{\pi}{2}}^{\frac{3\pi}{2}} \cos^2 \theta d\theta \\ &= \frac{1}{2} \int_{\frac{\pi}{2}}^{\frac{3\pi}{2}} \left(\frac{1 + \cos 2\theta}{2} \right) d\theta \\ &= \frac{1}{4} \int_{\frac{\pi}{2}}^{\frac{3\pi}{2}} (1 + \cos 2\theta) d\theta \\ &= \frac{1}{4} \int_{\frac{\pi}{2}}^{\frac{3\pi}{2}} d\theta + \frac{1}{4} \int_{\frac{\pi}{2}}^{\frac{3\pi}{2}} \cos 2\theta d\theta \\ u = 2\theta &\Rightarrow du = 2d\theta \Rightarrow d\theta = \frac{du}{2} \\ &= \frac{1}{4} \int_{\frac{\pi}{2}}^{\frac{3\pi}{2}} d\theta + \frac{1}{8} \int_{\pi}^{3\pi} \cos u du \\ &= \frac{1}{4} [\theta]_{\frac{\pi}{2}}^{\frac{3\pi}{2}} + \frac{1}{8} [\sin u]_{\pi}^{3\pi} \\ &= \frac{1}{4} \left[\frac{3\pi}{2} - \frac{\pi}{2} \right] + \frac{1}{8} [0 - 0] \\ &= \frac{\pi}{4} \end{aligned}$$

References



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